

Data Science

CSE558

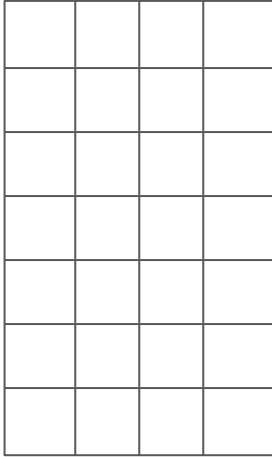
Recap

- Estimating Distinct Elements

Overview

- High dimensional data
- Dimensionality Reduction

Linear Algebra

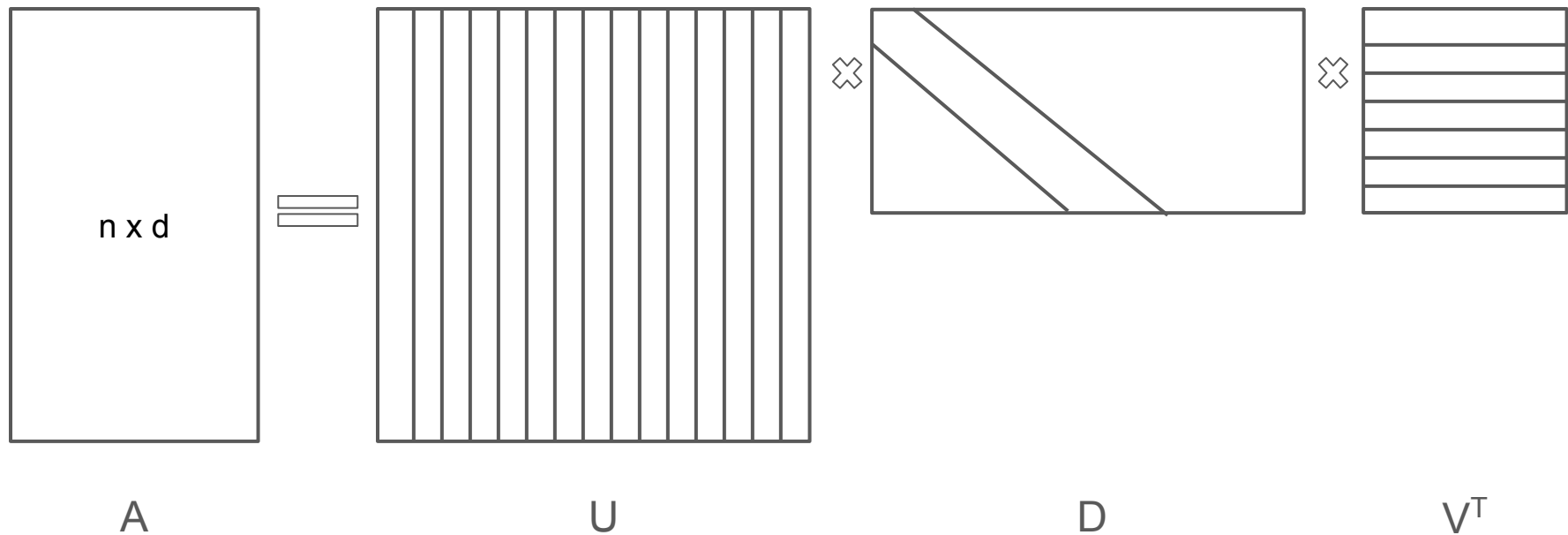


Rank-2

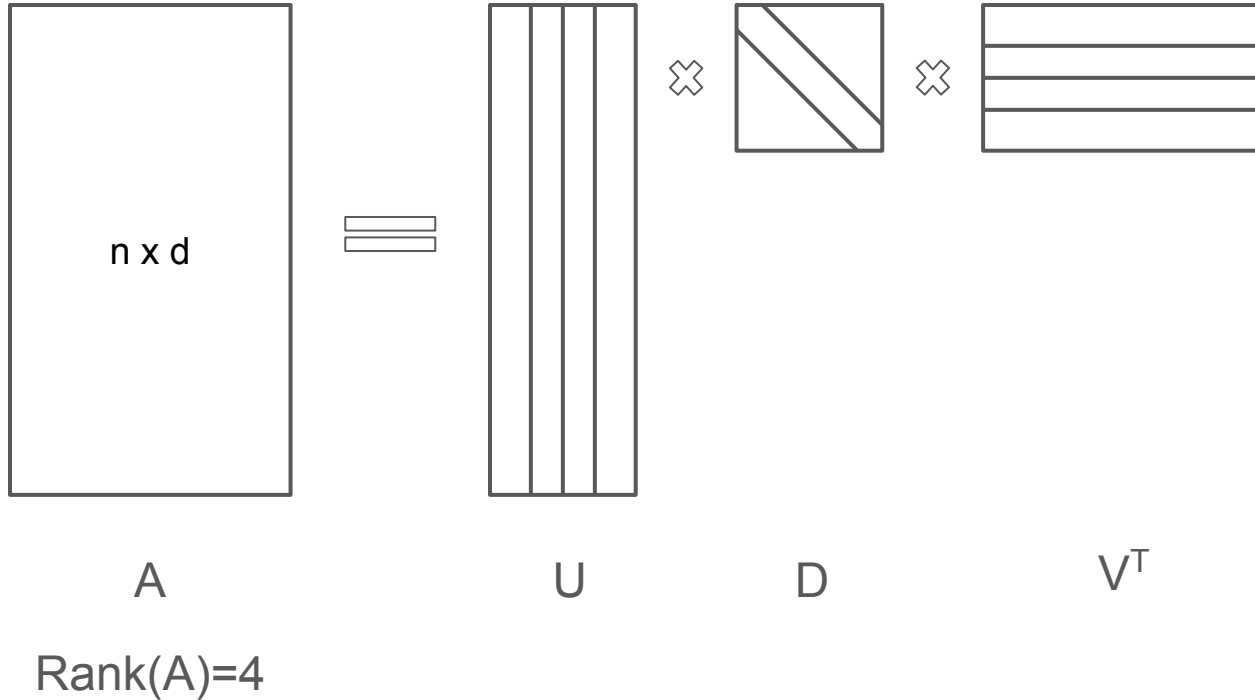
$$\alpha_1 \cdot \boxed{} + \alpha_2 \cdot \boxed{} = \boxed{}$$

Rank: Maximal number of linearly independent columns or rows

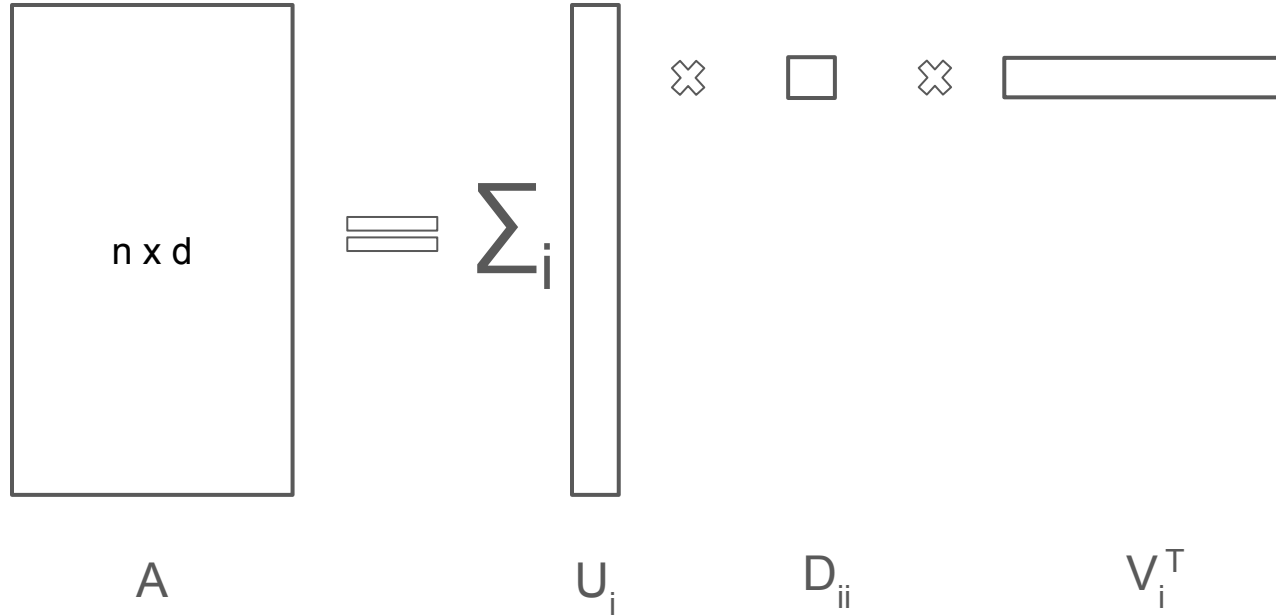
Singular Value Decomposition



Singular Value Decomposition



Singular Value Decomposition



Singular Vectors and Singular Values

$$\mathbf{v}_1 \in \operatorname{argmax}_{\|\mathbf{v}\|_2=1} \|A\mathbf{v}\|_2$$

$$\sigma_1(A) = \|A\mathbf{v}_1\|_2$$

$$\mathbf{v}_2 \in \operatorname{argmax}_{\mathbf{v} \perp \mathbf{v}_1; \|\mathbf{v}\|_2=1} \|A\mathbf{v}\|_2$$

$$\sigma_2(A) = \|A\mathbf{v}_2\|_2$$

Singular Vectors and Singular Values

Running time $O(nd \cdot \min(n,d))$

Compute, $B = A^T A$ and then compute eigen value decomposition of B .

Power Iteration:

Sample a random vector $\mathbf{x}_0$ and repeat following for k steps

$$\mathbf{x}_{t+1} = \frac{A\mathbf{x}_t}{\|A\mathbf{x}_t\|_2}$$

How to compute the next pair?

Projection

Projection of point $\mathbf{a}$ on a line passing through point $\mathbf{x}$ is $\left(\frac{\mathbf{x}\mathbf{x}^T}{\|\mathbf{x}\|_2^2} \right) \mathbf{a}$

Orthogonal distance $\left\| \left(\mathbf{I} - \frac{\mathbf{x}\mathbf{x}^T}{\|\mathbf{x}\|_2^2} \right) (\mathbf{a} - \mathbf{b}) \right\|_2^2$

Questions

Let A be an n by d matrix with singular vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$. For $1 \leq k \leq r$, let V_k be the subspace spanned by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$. For each k , V_k is the best k -dimensional subspace for A .

Let, $A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ then for any B ,

$$\|A - A_k\|_F \leq \|A - B\|_F$$

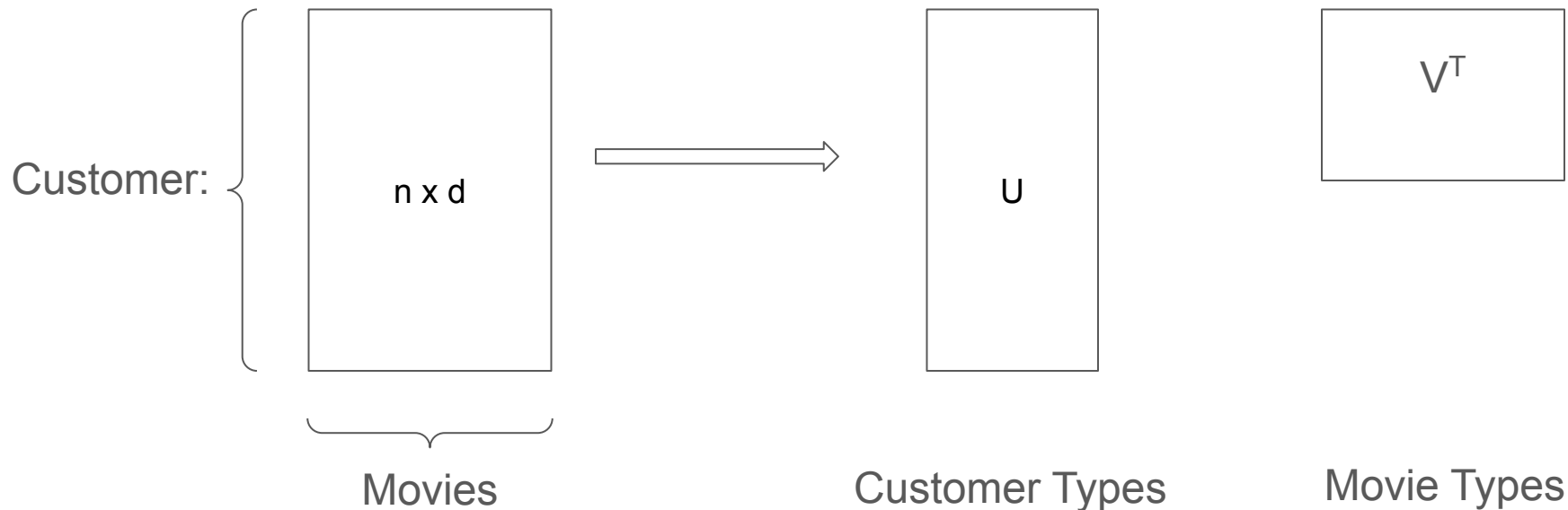
$$|A\mathbf{w}_1|^2 + |A\mathbf{w}_2|^2 \leq |A\mathbf{v}_1|^2 + |A\mathbf{v}_2|^2$$

Overview

- High dimensional data
- Dimensionality Reduction

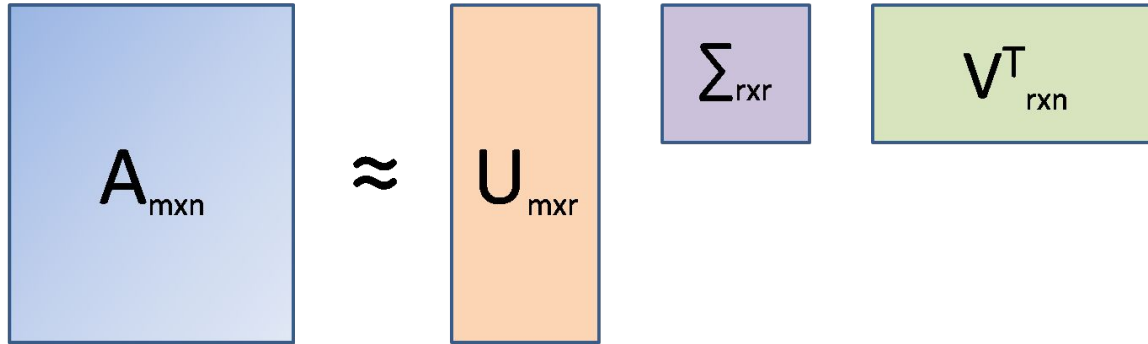
Dimensionality Reduction

Recommendation System (k types of movies):



Dimensionality Reduction

Latent semantic analysis

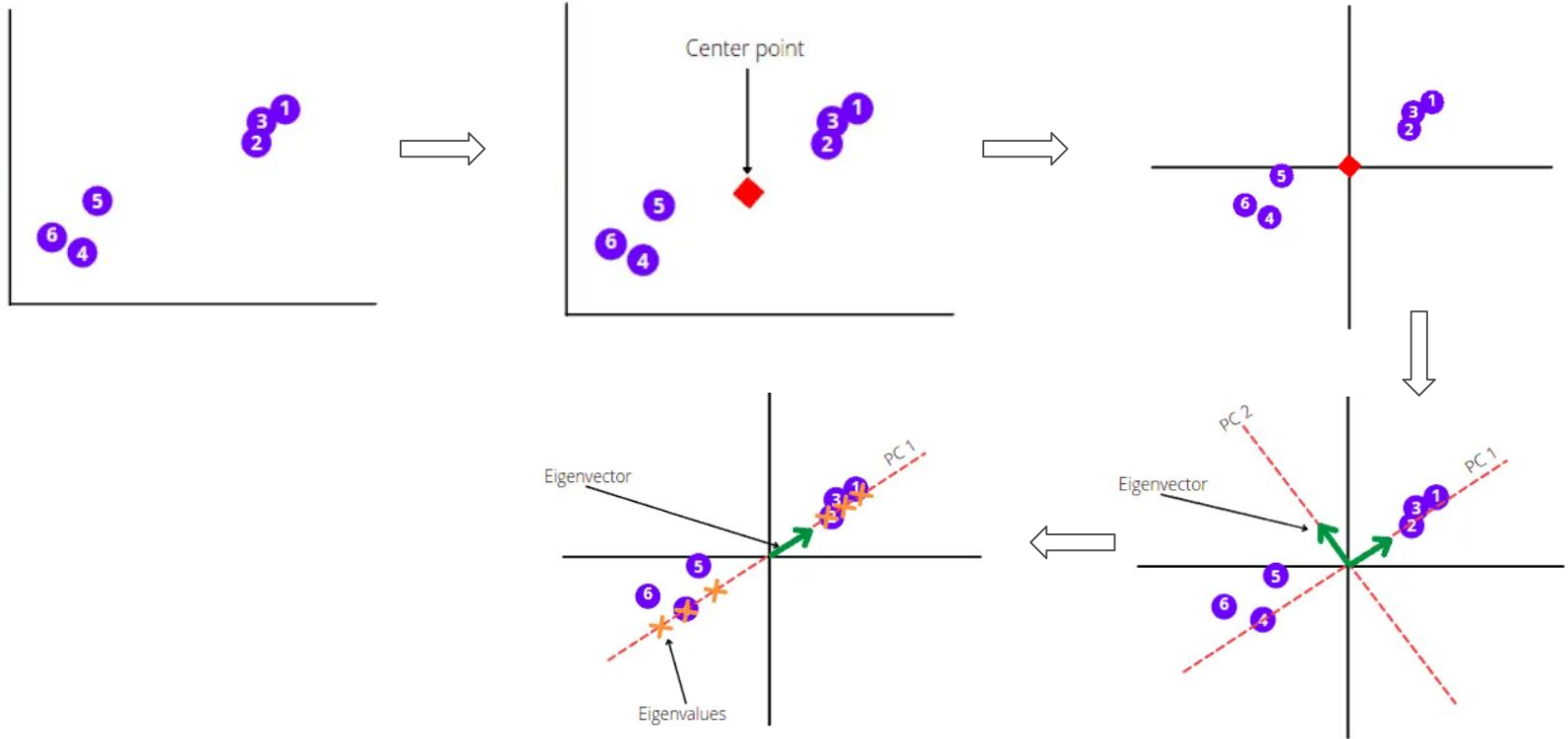


The diagram illustrates the matrix factorization process in Latent Semantic Analysis (LSA). It shows a large blue square representing matrix $A_{m \times n}$ on the left. To its right is an approximation symbol $\approx$. Further right is a tall orange rectangle representing matrix $U_{m \times r}$. To the right of U is a small purple square representing matrix $\Sigma_{r \times r}$. Finally, to the right of Σ is a green rectangle representing matrix $V^T_{r \times n}$. The boxes for U , Σ , and V^T are significantly smaller than the box for A , visually representing the dimensionality reduction from n to r latent dimensions.

$$A_{m \times n} \approx U_{m \times r} \Sigma_{r \times r} V^T_{r \times n}$$

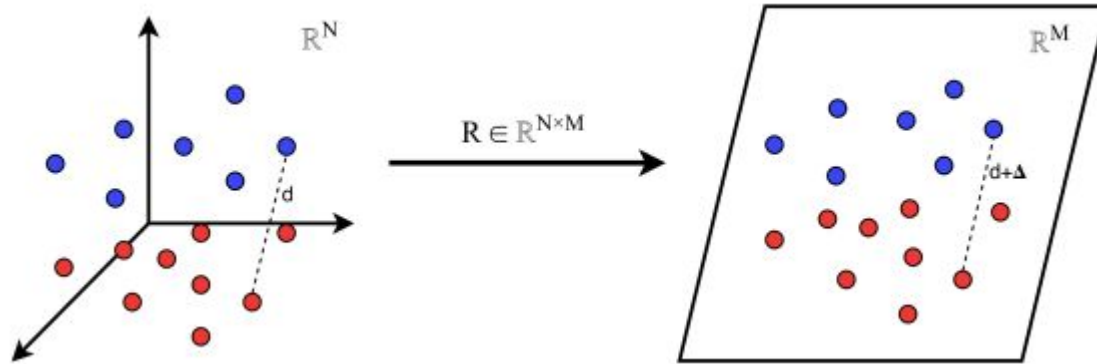
Application of SVD

PCA



Dimension Reduction

Projection



Simple Case

For any set of n points $(x_1, x_2, \dots, x_n)$ in a d -dimension. The goal is to preserve the following for every pair of points (x, y) and for some $\epsilon \in (0,1)$.

$$(1 - \epsilon) \|x_i - x_j\| \leq \|\tilde{x}_i - \tilde{x}_j\| \leq (1 + \epsilon) \|x_i - x_j\|$$

- Consider that the points are on 1st axis.
- Consider that the points are on a straight line.
- Consider that the points are on a plane (or 2-dim space).
- Consider that the points are on a k -dim space.